Lecture 2: LLM and its Application in Software Engineering

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Objectives

1. Understand what Large Language Models are and how they work
2. Use Cases in Software Engineering

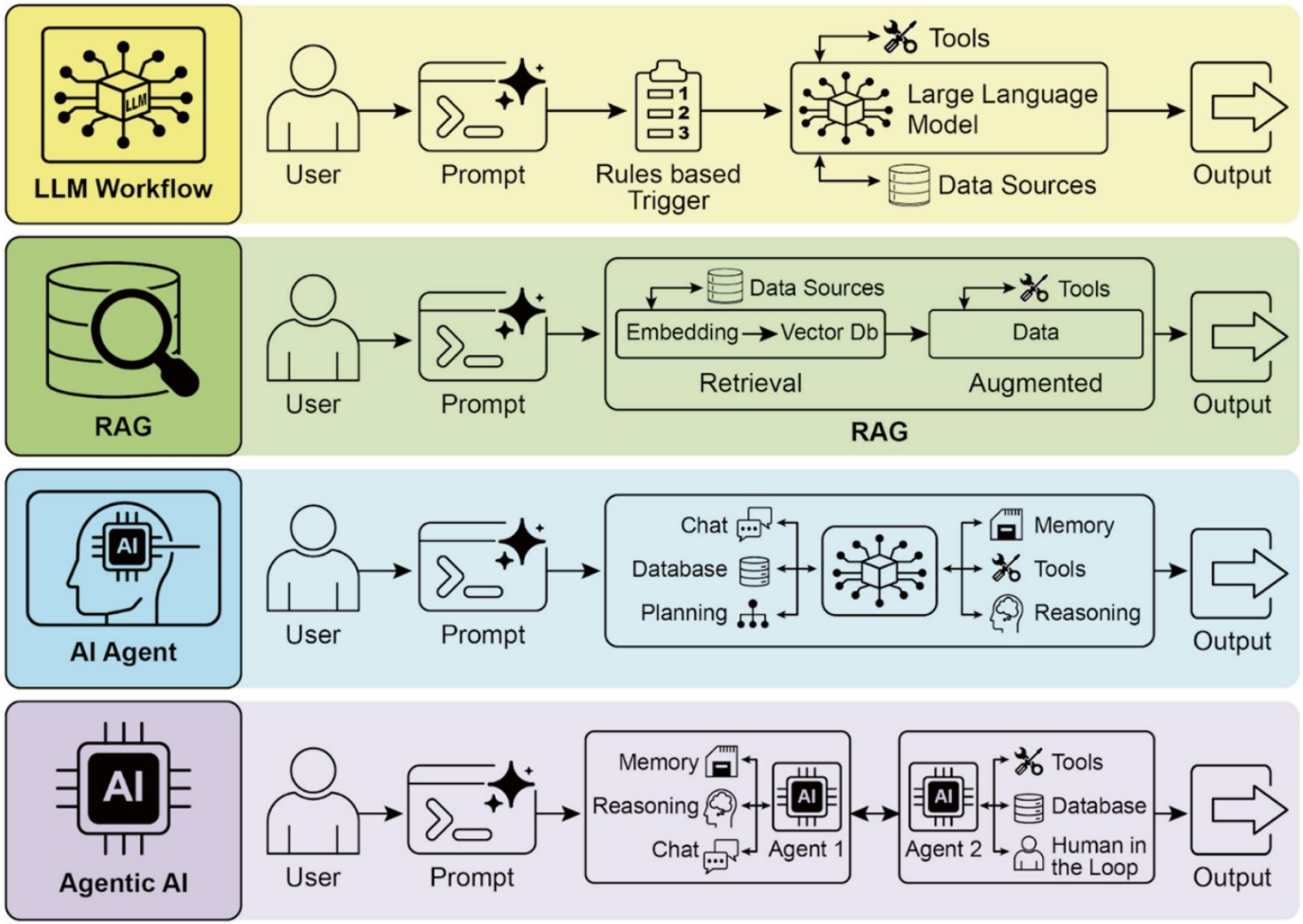
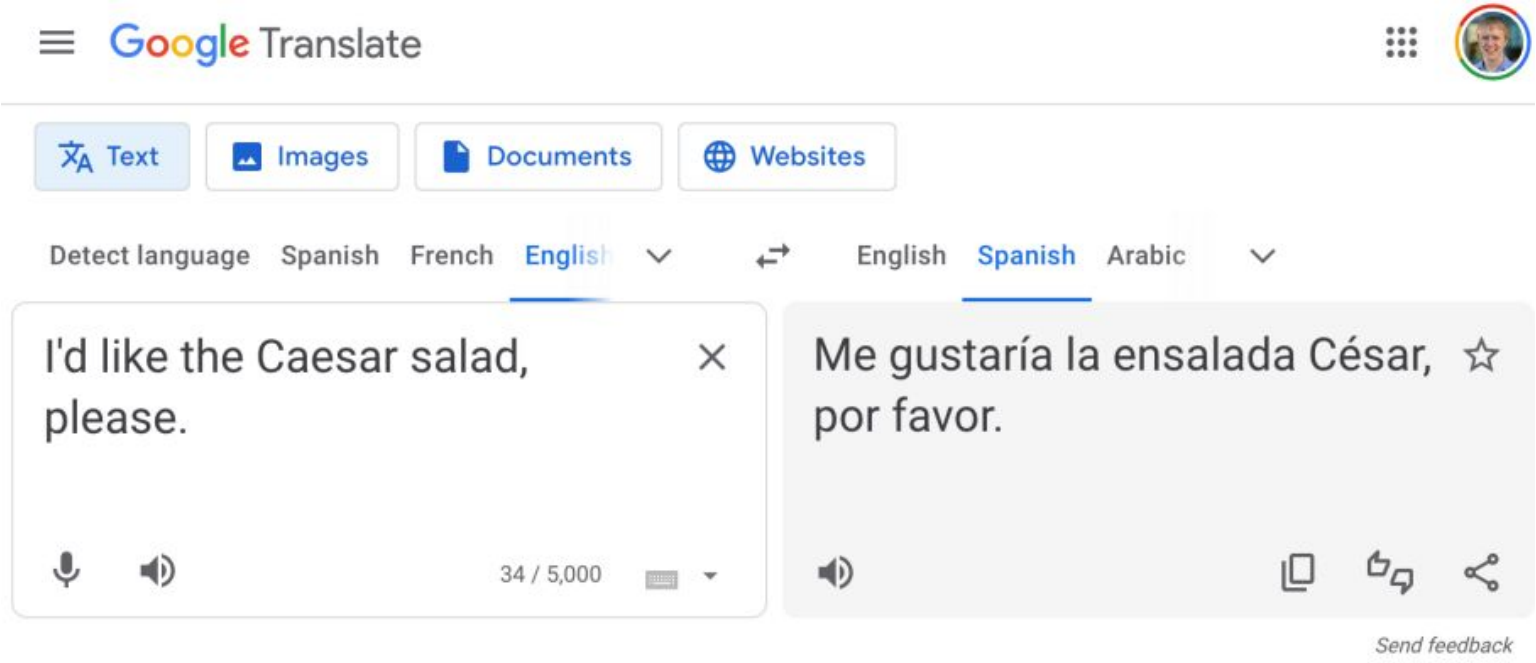


Fig. 2 Transitioning from LLMs to RAG, then to agentic RAG, and finally to agentic AI

In 2010-11 you could do this

In 2015-16 you could do this using Neural Network



Now, You Can Do This

Design an interactive programming learning system for kids with special needs

Designing an interactive programming learning system tailored for kids with special needs requires thoughtful integration of accessibility, engagement, and inclusivity. Here's a concept outline that balances interactive learning with the unique requirements of children with various abilities:

1. User Interface and Experience (UI/UX) Design

- **Customizable Layouts**: Allow users to modify fonts, color schemes, and text size to accommodate visual impairments or dyslexia. High-contrast themes should be available.
- **Simplified Navigation**: Use large buttons and intuitive icons to support children with motor skill challenges.
- **Text-to-Speech & Speech-to-Text**: Integrate tools that read instructions aloud and allow users to respond or code via voice.
- **Animations & Visuals**: Implement engaging visual prompts and animations that explain coding concepts in a fun, non-intimidating way.

2. Core Learning Modules

- **Gamified Learning**: Create games that teach fundamental programming concepts such as loops, conditionals, and variables. Games could involve character-based storytelling that reacts to user-coded inputs.
- **Adaptive Learning Paths**: Personalize lessons based on the child's pace and performance, ensuring each concept is revisited as needed without frustration.
- **Reward Systems**: Offer badges, points, and achievement systems for completed tasks to motivate continuous learning.

3. Accessibility Features

- **Multi-Sensory Input**: Support inputs beyond the keyboard, such as touch, speech, or eye-tracking technologies, making coding accessible for children with physical disabilities.
- **Audio Cues & Feedback**: Integrate auditory feedback when interacting with elements to reinforce learning for children with visual impairments.
- **Subtitles and Sign Language Options**: Include subtitles for audio elements and videos, and provide sign language support through animations or avatars for deaf children.

4. Interactive Coding Environment

- **Drag-and-Drop Interface**: Use a block-based coding system, similar to Scratch, where kids can drag and arrange code blocks to form programs. This reduces syntax errors and makes the learning process less intimidating.
- **Real-Time Error Guidance**: Implement a system that provides hints or visual cues when an error occurs, explaining the issue in simple language.
- **Live Output Panel**: Show immediate visual or audio results of the code they create, allowing for instant feedback and learning by trial and error.

5. Content Customization & Inclusivity

- **Diverse Characters & Stories**: Include characters that reflect different abilities and backgrounds to foster inclusivity.
- **Scenario-Based Learning**: Provide coding challenges based on relatable scenarios, such as coding for daily activities or problem-solving adventures, to enhance engagement.
- **Parent & Educator Tools**: Include dashboards where parents and teachers can monitor progress, adjust learning modules, and set specific goals or challenges.

6. Collaboration & Social Interaction

- **Pair Programming Mode**: Allow two kids to code together on shared projects. This feature can be adjusted for remote interaction, encouraging teamwork and social skills.
- **Community Sharing**: Provide a safe space for kids to share their projects, view others' work, and gain inspiration while promoting positive social interaction and feedback.
- **Mentorship Programs**: Connect kids with mentors who can assist with learning through video calls or chat support, fostering a guided learning experience.

7. Assistive Tutorials & Support

- **Interactive Walkthroughs**: Use animated guides or characters that lead children through coding exercises step-by-step.
- **Comprehensive Help Center**: Offer a support section with video tutorials, FAQs, and troubleshooting tips that are accessible and easy to understand.

7. Assistive Tutorials & Support

- **Interactive Walkthroughs**: Use animated guides or characters that lead children through coding exercises step-by-step.
- **Comprehensive Help Center**: Offer a support section with video tutorials, FAQs, and troubleshooting tips that are accessible and easy to understand.
- **Parent/Educator Resources**: Provide printable materials, tips for effective learning strategies, and suggestions for adaptations specific to different needs.

8. Security & Privacy

- **Safe Environment**: Implement secure logins and user data protections, ensuring parents have full control over who can interact with their child's profile.
- **Customizable Access**: Allow parents and educators to set content limitations and control communication features for a secure and focused learning space.

9. Rewards Beyond the Screen

- **Physical Kits**: Include coding kits that interface with the digital platform, allowing kids to see their code affect tangible outcomes (e.g., programmable robots or lights).
- **Printable Achievements**: Certificates and other printed materials to celebrate milestones and maintain a sense of accomplishment.

This system would cater to the varying needs and abilities of kids while creating an engaging, empowering, and fun learning environment where they can develop foundational programming skills tailored to their pace and preferences.

What are Large Language Models?

- LLMs predict the next word like how humans construct sentences

Given Sequence: "Once upon a"
Human Prediction: "*time*" (common phrase)
Given Sequence: "I need to buy some"
Human Prediction: "*groceries*" (everyday context)

What are Large Language Models?

Input: Sequence of words

Output: next word

3 steps:

- Analyzes the context and previous words
- Computes probabilities for possible next words
- Chooses the word with the highest probability

What are Large Language Models?

- Given Sequence: "Mary had a __"
- Prediction: __

What are Large Language Models?

- Given Sequence: "Mary had a __"
- Prediction:
  - "little" (highest probability)
  - Alternatives: "red", "delicious", etc.

Next Token	Probability
red	0.2
little	**0.4**
delicious	0.2
...	...

What are Large Language Models?

- Given Sequence: "Mary had a little __"
- Prediction:
  - "lamb" (highest probability)
  - Alternatives: "mouse", "kettle", etc.

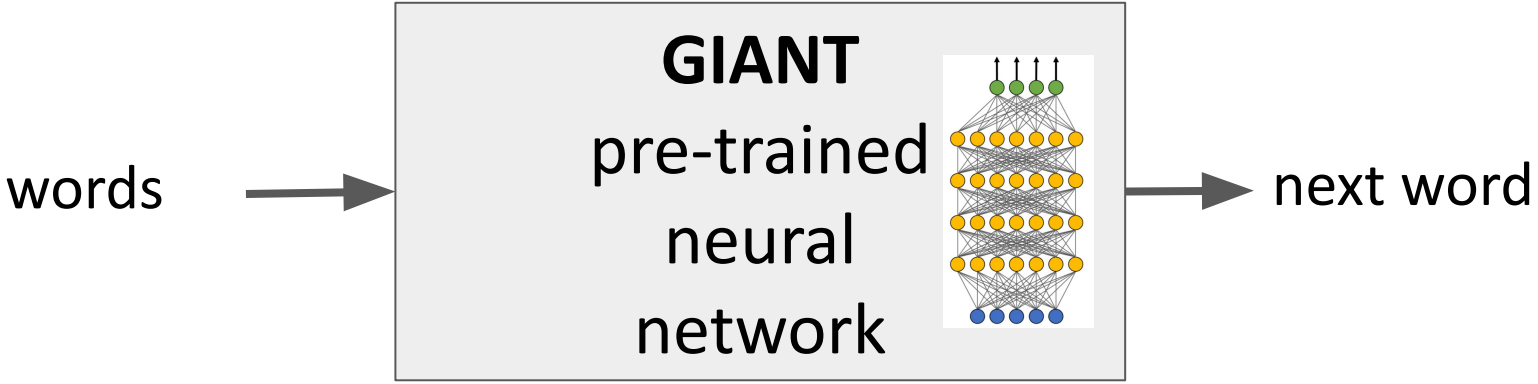
Next Token	Probability
mouse	0.4
lamb	**0.3**
Kettle	0.2
...	...

What are Large Language Models?

- LLMs predict the next word, given a sequence of words
  - *p(“word” | sequence of words before it)*
  - e.g., n-gram
- p(“soon”| “see”,“you”) > p(“puppy” | “see”,“you”)

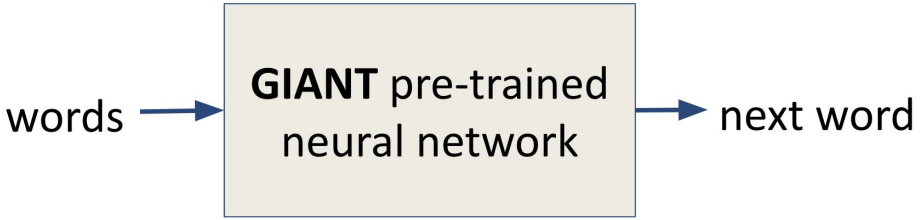
What are Large Language Models?

Adding the necessary complexity…



What are Large Language Models?

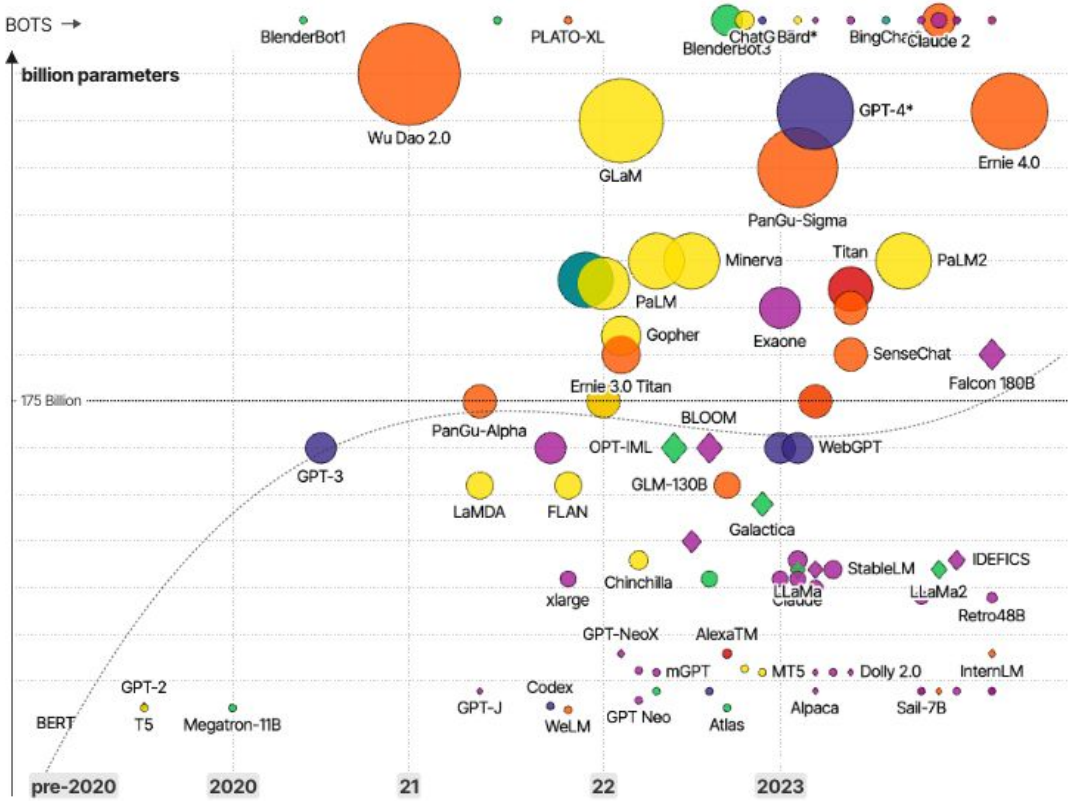
- *Giant*: billions of parameters
- *Pre-trained*: unsupervised learning over very large datasets (books, Internet, etc.)
- *Neural network*: type of machine learning model inspired by human brain



The Rise and Rise of A.I. Large Language Models (LLMs) & their associated bots like ChatGPT

size = no. of parameters open-access

Amazon-owned Chinese Google Meta / Facebook Microsoft OpenAI Other



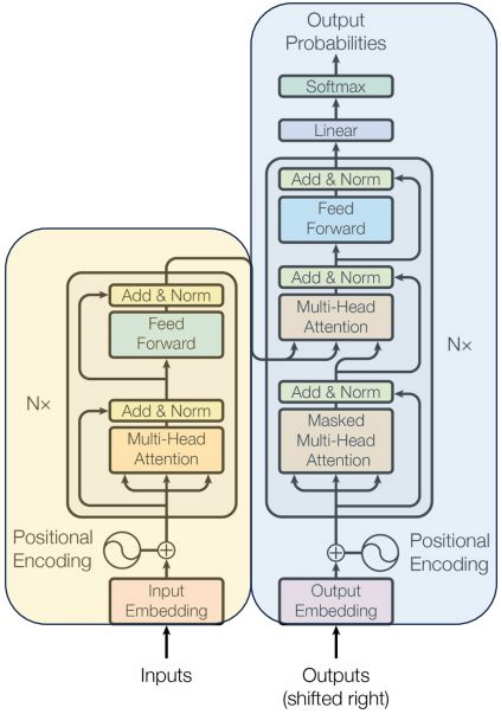
David McCandless, Tom Evans, Paul Barton
Information is Beautiful // UPDATED 2nd Nov 23

source: news reports, LifeArchitect.ai
* = parameters undisclosed // see the data

It Started in 2018..

Attention Is All You Need:
https://user.phil.hhu.de/~cwurm/wp-content/uploads/2020/01/7181-attention-is-all-you-need.pdf

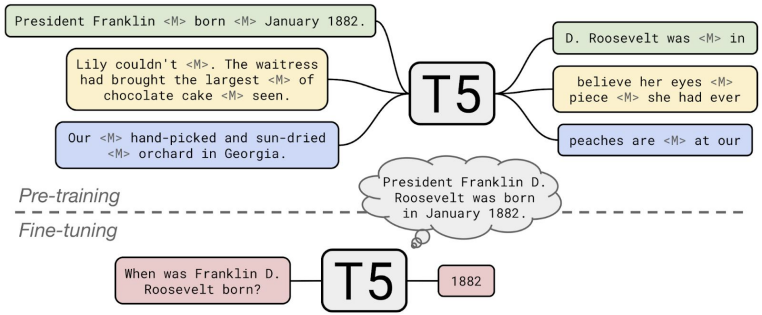
BERT (OCT 2018)
Parameters:
0.108B (base ver.)
Representation



GPT (June 2018)
Parameters:
0.110B
Generation

Generalizing to Unseen Tasks

- LLMs can be used for different tasks by pre-training a “base” model and then fine-tuning for the task(s) of interest
  - Bug detection
  - Sentiment analysis
  - Text summarization
  - Question answering
  - Code completion



Question Answering with T5

Generalizing to Unseen Tasks

- Can we do better? Just train once and that works for every task?
- Multi-task Training:
  - Humans don’t need such large volumes of data to learn
  - Can the models do the same?

Turns out It Can!

- Train a model that can perform NLP tasks in a zero-shot manner
  - One of such first models: **GPT-2**
  - Parameters: 1.542B

Single Task Model

P(output | input)

General Model

P(output | input, *task*)

GPT Continue to get Bigger and Bigger

- GPT: 0.110B (2018)
- GPT-2: 1.542B (2019)
- GPT-3: 175.0B (2020)
- GPT-3.5 aka ChatGPT (2022)
  - GPT-3 can solve problems with zero-shot, one-shot, and few-shot manner
  - It did reasonably well on tasks it has *never even seen*, which lead to further improvement in 2020 - GPT-3.5, that we know as **ChatGPT**

LLM - Architecture (Summary)

- Encoder-only (BERT)
  - Great for classification tasks, but hard to do generation
- Decoder-only (GPT)
  - Better generalization after pre-training
- Encoder-decoder (T5)
  - Good for tasks like translation, summarization

Transformer models: an introduction and catalog: https://arxiv.org/pdf/2302.07730

LLM Limitations

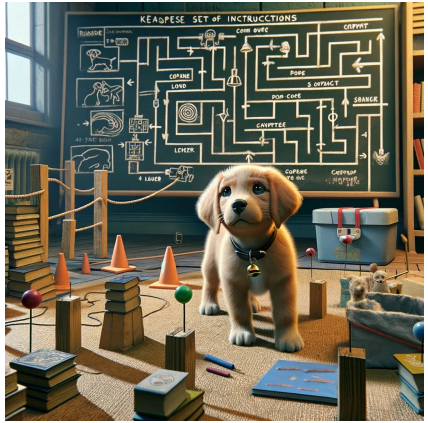
LLM Limitations

- Price/resources
- Potential bias
- Token limits
- Randomness
- Hallucinations

Other caveats of using LLMs

- Ethical considerations (bias, privacy)
- Technical challenges (integration complexity, resource demands)
- Randomness and unpredictability of outputs
- Ongoing maintenance and updates
  - LLMs change quickly

Questions



Reading

- Large Language Model-Based Agents for Software Engineering: A Survey: https://arxiv.org/pdf/2409.02977 - Dec 2025
- From LLMs to LLM-based Agents for Software Engineering: A Survey of Current, Challenges and Future: https://arxiv.org/pdf/2408.02479 - April 2025

Generative AI in Real World Project Development

Generative AI in Software Development Market

Size, by application, 2022-2032 (USD Billion)

The Market will Grow At the CAGR of: **21.4%** The forecasted market size for 2032 in USD: **$169.2M**

MarketResearch.biz
WIDE RANGE OF GLOBAL MARKET REPORTS

Responsible Practices

- Understand the Limitations
- Verify the Output
- Respect Privacy and Confidentiality

Recommendations

- Start Small
- Collaborate and Discuss

What can We Do?

- Using LLMs in test automation
- Converting requirements to machine readable formats
- Auto code completion
- Code comment generation
- Program repair
- Code review

What else?

Requirement Engineering

- estimation
- Generating SRS from interview transcripts
- Creative Requirements generation
- Requirements Analysis
- requirements traceability
- Improving quality and completeness of Requirements
- Ambiguity Management
- Requirements Generation
- Identifying Ambigous and Conflicting Requirements
- generating code based on requirements
- Process key information from complex meetings
- Model Checking

Software Design

- Model Extraction
- Suggest design patterns
- Generating system architecture model
- Simplification of architecture
- Language translation
- migrating monolith to microservices

Software Implementation

- Code Generation
- generate code patches
- pseudo code reasoning
- reduction of the legacy code (can be rewritten faster, so no need to maintain suboptimal solutions)
- taking on the repetitive simple code writing (e.g. application development)
- Just in time developer feedback
- reverse engineering

Quality Assurance

- Acceptance Criteria Generation
- Test Cases Generation
- test cases maintenance, gaps identification
- tight connection between code documentation and test cases
- Code review
- test case generation
- Code Analysis
- Model Extraction

Deployment/Maintenance

- Bug Detection and Fixing
- code explaination and documentation
- security vulnerabilities detection
- Code Refactoring
- optimized resource allocation (e.g. recommendations on AWS configs)
- LLMs-driven DevOps

Project Management

- more transparency (easier documentation, less admin work)
- Iterative process improvement (assisting with identification of weaknesses and suggestions about how to fix them)
- Project Planning and Task Prioritization
- suggest project plan and task assignment

Software Processes

- generate new retro games
- Help with Scrum Meetings (Scrum Master Assistant)
- Story point estimation
- Agile retros (scrum master)
- retrieve knowledge from GIT repos
- Pair programing practice

Professional practices

- cheap advisor always available
- Psychological aspects
- Retraining professionals
- Replacement of people/teams downsizing

Tools

- Aid code commit, pull push request
- shell command and scripting
- New development environment

Education

- personalized learning
- exercises and grading

Possible Use Cases …

- A developer uses an LLM to find vulnerabilities in code, aiming to catch issues missed by static analysis tools
- A developer uses an LLM to generate code for parsing structured input and performing numerical analysis
- A tester uses an LLM to create functional test cases from a requirements document
- A developer uses an LLM to generate software documentation from code, summarizing functionality and interface
- A software engineer unfamiliar with SQL uses an LLM to generate a SQL query from a natural language description
- A developer uses an LLM to translate C++ classes to Rust, expecting the translated code to pass tests and be more secure and memory efficient
- A developer uses an LLM to generate synthetic test data for a new feature, expecting correct and representative data
- A developer provides an LLM with failing production code and failure details, expecting help diagnosing and fixing the issue
- A software architect uses an LLM to check if deployment-ready code aligns with system architecture, expecting reliable detection of deviations

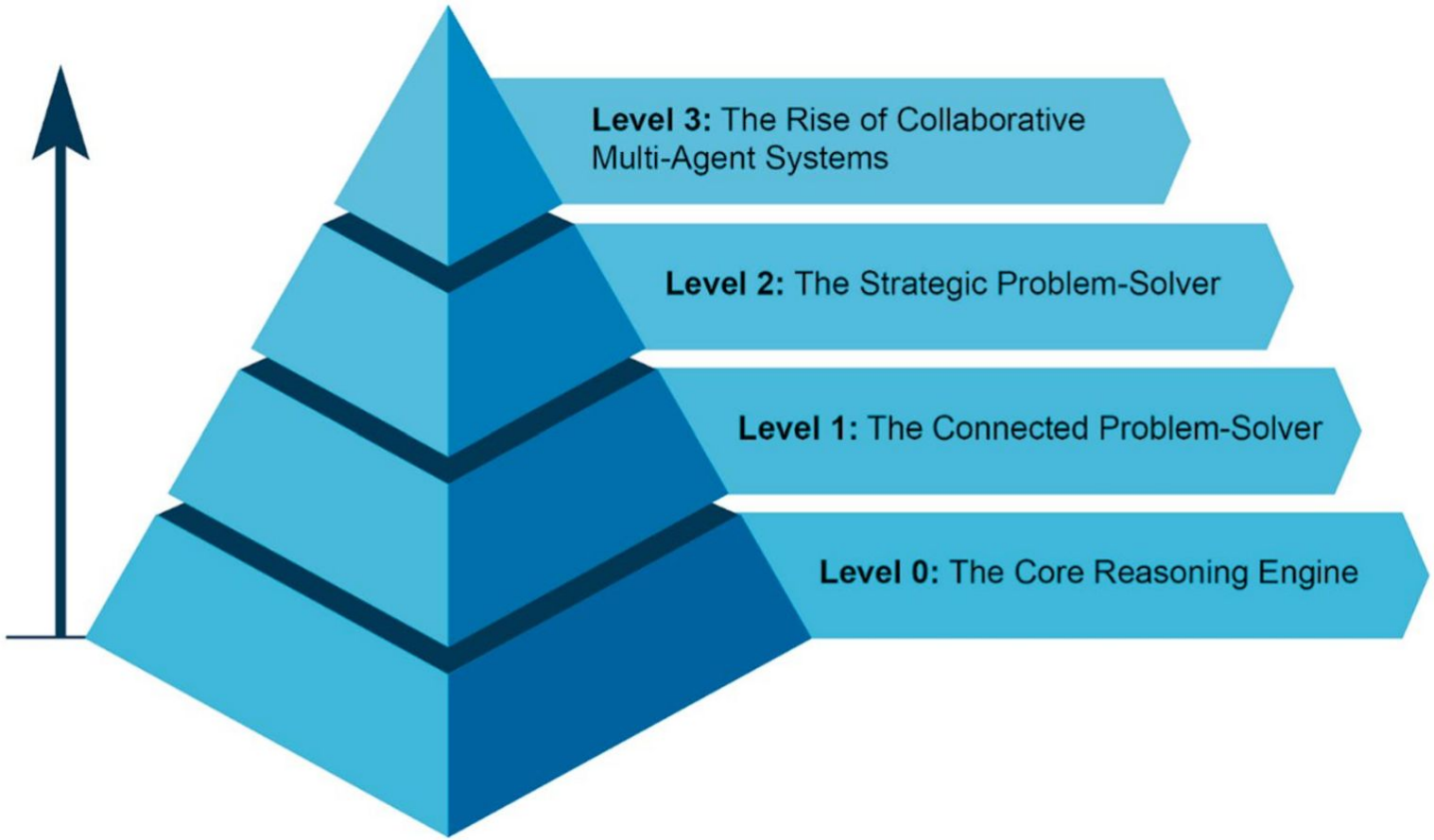


Fig. 3 Various instances demonstrating the spectrum of agent complexity

Trends: Industry Evolution

- Automated code review
- AI-driven architecture
- Intelligent debugging
- Self-modifying code

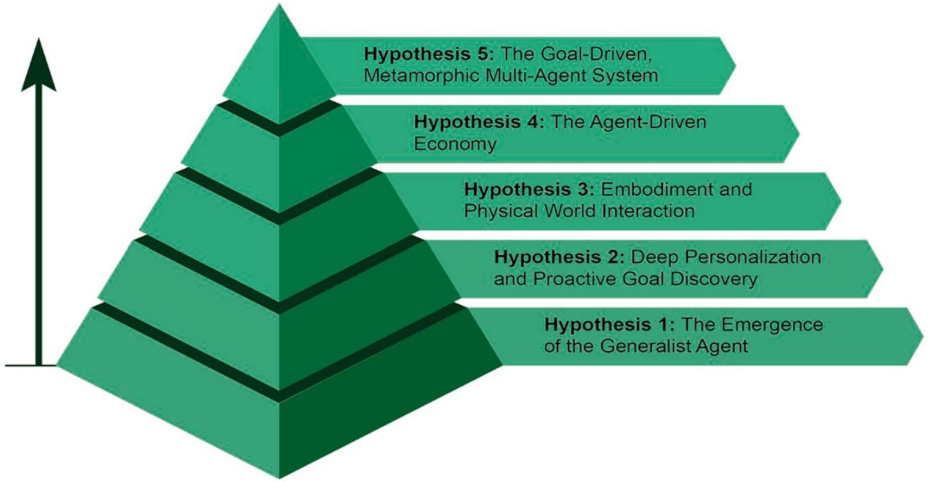


Fig. 4 Five hypotheses about the future of agents

Trends: Challenges

- Intellectual Property (IP) Concerns
- Privacy
- Explainability
- Security

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AI

Anthropic cut up millions of used books to train Claude — and downloaded over 7 million pirated ones too, a judge said

What data did Vyro AI leak?

- AI prompts that users typed into the apps
- Bearer authentication tokens
- User agents

Generative AI Data Policy Violations Average 223 Per Month

This increase in data being sent to AI tools creates additional security risks for organizations; according to Netskope the number of known data policy violations as a result of employees using

Future Trends: Predictions

- Will NOT replace the need for software developers
- Free developers to do more of the fun and/or more critical software engineering activities.
- Speed up digital transformation in traditional sectors by increasing access to organizations lagging in digitization
- Democratize software
- *Personalized software*: You would build your own little softwares that you can run locally

Career Impact: Required Skills

- Effective Prompt engineering
- AI-Pair programmer
- Collaboration and Team Dynamics
- AI tool evaluation
- Output validation (Benchmarking)
- System integration

Career Impact: Job Market Changes

- New roles emerging (e.g., AI Engineer)
- Skill requirements shifting
- Tool proficiency important
- Collaboration focus

Questions?

Questions

Q: If you want to develop one agent to help you in development, what would be the purpose of the agent?

Summary: Benefits of LLMs

- Enhanced user experience and engagement
  - (e.g., virtual assistants, personalization)
- Increased efficiency and automation
  - (e.g., programming assistants)
- Data-driven insights and decision-making
  - (e.g., processing large volumes of user product reviews)

Summary: However..

- LLMs are trained on the Internet
  - The Internet is not always a fair and happy place :-)
- LLMs are not smart
  - They are word guessers and will comply with some bad prompts
- LLMs can make mistakes
  - Wrong answers and hallucinations are possible